



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

2002
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time — 5 minutes
- Working time — 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks — 120

- Attempt questions 1–8
- All questions are of equal value, the mark value is shown beside each part.

Examiner: D.M.Hespe

Note: This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 120

Attempt Questions 1–8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Find

(i) $\int \sin^{-1} x \, dx$ 2

(ii) $\int \frac{x}{1+x^4} \, dx$ 2

(iii) $\int \tan^3 x \, dx$ 2

(b) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\cos x}$ using the substitution $t = \tan \frac{x}{2}$. 3

(c) Given that $I_n = \int_1^e (\ln x)^n \, dx$, $n = 0, 1, 2, \dots$, 3
show that $I_n = e - nI_{n-1}$.

(d) If $x = \frac{\pi}{4} - u$,

(i) Show that $\tan x = \frac{1 - \tan u}{1 + \tan u}$. 1

(ii) Hence or otherwise, show that $\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) \, dx = \frac{\pi}{8} \ln 2$. 2

Question 2 (15 marks) Use a SEPARATE writing booklet.

- (a) Explain the flaw in this “proof” that $i = -i$.

2

$$\begin{aligned}
 i &= i \\
 \sqrt{-1} &= \sqrt{-1} \\
 \sqrt{\frac{-1}{1}} &= \sqrt{\frac{1}{-1}} \\
 \frac{\sqrt{-1}}{\sqrt{1}} &= \frac{\sqrt{1}}{\sqrt{-1}} \\
 \frac{i}{1} &= \frac{1}{i} = \frac{-(-1)}{i} = \frac{-i^2}{i} = -i \\
 \therefore i &= -i
 \end{aligned}$$

- (b) $u = -3 - 4i$ and $v = 1 - i$ are two complex numbers. Express in the form $x + iy$, where x and y are real:

(i) $\bar{u} - v$

1

(ii) $\frac{2u}{v}$

2

(iii) $\sqrt{u}$

2

- (c) On an Argand diagram sketch the region defined by

2

$$\{z: -\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{6} \cap |z| \leq 1\}.$$

- (d) (i) If a, b are the complex numbers represented by the points A and B on an Argand diagram, what geometrical properties correspond to the modulus and argument of $\frac{b}{a}$?

2

- (ii) Show that, if the four points representing the complex numbers $z_1, z_2,$

4

$z_3,$ and z_4 are concyclic, the fraction $\frac{(z_1 - z_2)(z_3 - z_4)}{(z_3 - z_2)(z_1 - z_4)}$ must be real.

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Question 3 (15 marks) Use a SEPARATE writing booklet.

- (a) Reduce to irreducible factors over the complex field: $x^3 - 4x^2 + 7x - 6$. 3
- (b) Find the polynomial $f(x)$ of the fourth degree such that $f(0) = f(1) = 1$, $f(2) = 13$, $f(3) = 73$ and $f'(0) = 0$. 4
- (c) (i) Prove that if $P(x)$ has a root of multiplicity m , then $P'(x)$ has a root of multiplicity $m - 1$. 2
- (ii) Find the value of c if the polynomial $5x^5 - 3x^3 + c$ has a positive repeated root. 3
- (d) Let α, β, γ be the roots of the equation $x^3 + px + q = 0$, where $q \neq 0$. Find, in terms of p and q , the coefficients of the cubic equation whose roots are α^{-1}, β^{-1} , and γ^{-1} . 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) Solve the simultaneous equations: 4
- $$x^2 + xy + y^2 = 7,$$
- $$2x^2 - xy + y^2 = 28.$$
-
- (b) Show that if $b^2 < 4ac$, the value of the function $ax^2 + bx + c$ will have the same sign as a for all real values of x . 2
-
- (c) (i) By considering the expression $x^2 - 2xy + 5y^2 + 2x - 14y + k$ as a quadratic function of x , show that it is positive for all real values of x and y if $k > 10$. 4
-
- (ii) Show that if $k = 10$, the expression may be written in the form $(x + py + q)^2 + (ry + s)^2$, and hence find the simultaneous values of x and y for which the expression is zero. 4
-
- (iii) Deduce the minimum value of the expression for a general value of k . 1

Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle P of mass m starts from rest at a point O and falls under gravity in a medium where the resistance to its motion has magnitude mkv , v being the speed of the particle and k is a constant.

(i) Draw a diagram to show the *forces* acting on the particle during this downward path, and hence write down the equation of motion. 2

(ii) Show that the expression for its velocity v at any time t is given by 2

$$v = \frac{g}{k}(1 - e^{-kt}).$$

(iii) Explain what is meant by the *terminal velocity* and find an expression for the terminal velocity V_T . 3

- (b) A second particle Q , also of mass m , is fired vertically upwards from O with initial speed u , so that P and Q leave O simultaneously.

(i) Draw a diagram to show the *forces* acting on the particle during this ~~downward~~ path, and hence write down the equation of motion. 2

upward.

(ii) Find an expression for the time t when Q comes to rest. 3

(c) Show that, at the instant Q comes to rest, the velocity of P is given by: 3

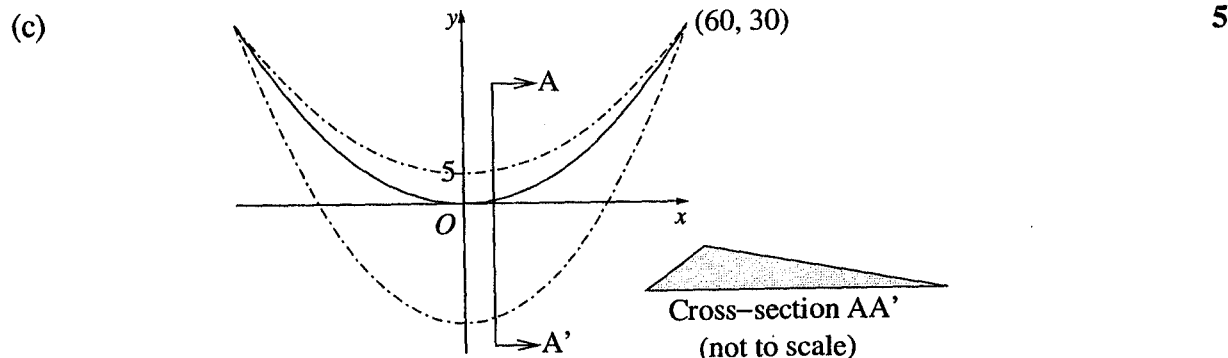
$$v = \frac{V_T u}{V_T + u}.$$

Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that if x is real, the expression $\frac{(x-2)^2}{x-1}$ cannot take any value between -4 and 0 . 2
- (ii) Sketch the graph of the expression. 3
- (iii) Show that the equation $\frac{(x-2)^2}{x-1} = \frac{k}{x}$ has three real roots if k is positive, but only one real root if k is negative. 2
- (b) For $z = r(\cos \theta + i \sin \theta)$, find r and the smallest positive value of θ which satisfy the equation $2z^3 = 9 + 3\sqrt{3}i$. 2
- (c) Using the method of shells find the volume of the solid formed when the region bounded by the curve $y = x^2 + 1$ and the x -axis between $x = 0$ and $x = 2$ is rotated about the y -axis. 3
- (d) Explain why, if $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \lim_{n \rightarrow \infty} \left(n \times \sqrt{1 + \frac{1}{n}} - n \right)$, then the limit is not zero, but a half. 3

Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the rational roots of $x^4 + 2x^3 - 17x^2 - 18x + 32 = 0$ using the substitution $y = x^2 + x$, or otherwise. 2
- (b) (i) Prove that the medians of a triangle are concurrent at a point which is a point of trisection of each median. [A *median* of a triangle is a line from a vertex to the mid point of the opposite side.] 3
- (ii) If the medians of triangle ABC meet at G , and AG is produced to K so that $AG = GK$, prove that the triangle BGK is similar to the triangle whose sides are equal in length to the three medians. 3
- (iii) Also show that the area of the triangle whose sides are equal in length to the medians is $\frac{3}{4}$ of the area of triangle ABC . 2



Barcan sand dunes are parabolic in plan view and are triangular in cross section with the inner face having an angle of repose of $\tan^{-1} \frac{3}{4}$ to the horizontal and the outer face at $\tan^{-1} \frac{1}{6}$ to the horizontal. The figure above shows one such dune (dimensions are in metres). Calculate the volume of sand.

Question 8 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) A circular disc, centre A , of radius a , rolls without slipping along the axis of x . Initially the point P on the edge of the disc is at the origin of coordinates. Prove that, when the radius AP has turned through an angle θ , the coordinates of P are: $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$. 3

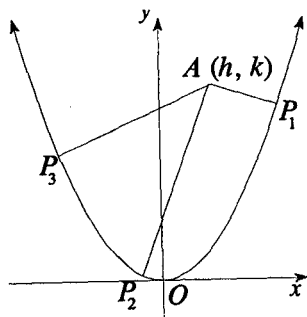
- (ii) The length, ℓ , of a curve, $y = f(x)$, is given by 3

$$\ell = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

When P is again in contact with the axis of x , prove that the length of its path is $8a$.

- (b) Sum the series, n being a positive integer, 4
- $$1 + x \cos x + x^2 \cos 2x + x^3 \cos 3x + \dots + x^n \cos nx.$$

- (c) (i) 3



Prove that, in general, three normals can be drawn from any point to a parabola.

- (ii) Also show that if P_1 , P_2 , and P_3 have coordinates (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) respectively, then $x_1 + x_2 + x_3 = 0$. 2

End of paper

$$\begin{aligned}
 \text{(a) (i)} \quad \int \sin^{-1} x \, dx &= \int \sin^{-1} x \cdot \frac{d(x)}{dx} dx \\
 &= x \sin^{-1} x - \int x \cdot \frac{1}{\sqrt{1-x^2}} dx \\
 &= x \sin^{-1} x + \sqrt{1-x^2} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad I &= \int \frac{x}{1+x^4} dx \\
 \text{let } u &= x^2 \Rightarrow du = 2x dx
 \end{aligned}$$

$$\begin{aligned}
 \therefore I &= \frac{1}{2} \int \frac{du}{1+u^2} \\
 &= \frac{1}{2} \tan^{-1} u + C \\
 &= \frac{1}{2} \tan^{-1}(x^2) + C
 \end{aligned}$$

$$\text{(iii)} \quad I = \int \tan^3 x \, dx = \int \tan x \cdot \tan^2 x \, dx$$

$$\begin{aligned}
 \text{ie } I &= \int \tan x (\sec^2 x - 1) \, dx \\
 &= \int \tan x \sec^2 x \, dx - \int \tan x \, dx \\
 &= \frac{1}{2} \tan^2 x - \log_e (\cos x) + C \\
 &= \frac{1}{2} \tan^2 x + \log_e (\sec x)
 \end{aligned}$$

$$\begin{aligned}
 \int_0^1 \frac{dx}{1+\cos x} & \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} \\
 &= \int_0^1 \frac{2 \, dt}{\left(1 + \frac{1-t^2}{1+t^2}\right)(1+t^2)} \quad \frac{dx}{dt} = \frac{2}{\sec^2 \frac{x}{2}} \\
 &= 2 \int_0^1 \frac{dt}{2} \quad \frac{dx}{dt} = \frac{2}{1+t^2} \\
 &= \left[t \right]_0^1
 \end{aligned}$$

$$\begin{aligned}
 &\text{When } x=0, t= \\
 &\text{When } x=\frac{\pi}{2}, t=
 \end{aligned}$$

$$= 1 - \frac{1}{2}$$

$$\begin{aligned}
 \text{(c)} \quad I_n &= \int_1^e (\ln x)^n \, dx \\
 &= \int_1^e (\ln x)^n \cdot \frac{d(x)}{dx} dx \\
 &= \left[x \ln x \right]_1^e - \int_1^e x \cdot n (\ln x)^{n-1} \cdot \frac{1}{x} dx \\
 &= e - n \int_1^e (\ln x)^{n-1} dx
 \end{aligned}$$

$$\therefore I_n = e - n I_{n-1}$$

$$\begin{aligned}
 \text{(d)} \quad \text{(i)} \quad \tan\left(\frac{\pi}{4} - u\right) &= \frac{\tan \frac{\pi}{4} - \tan u}{1 + \tan \frac{\pi}{4} \tan u} \\
 &= \frac{1 - \tan u}{1 + \tan u}
 \end{aligned}$$

$$\text{(ii)}$$

(b)

Let $z = x \operatorname{cis} \theta$, for real x

$$1 + z + z^2 + z^3 + \dots + z^n = \frac{z^{n+1} - 1}{z - 1}$$

$$\operatorname{Re}(1 + z + z^2 + z^3 + \dots + z^n) = 1 + x \cos \theta + x^2 \cos 2\theta + \dots + x^n \cos n\theta$$

(De Moivre's Theorem)

$$\therefore 1 + x \cos \theta + x^2 \cos 2\theta + \dots + x^n \cos n\theta = \operatorname{Re} \left(\frac{z^{n+1} - 1}{z - 1} \right) \quad \checkmark$$

So

$$\frac{z^{n+1} - 1}{z - 1} = \frac{x^{n+1} \operatorname{cis}(n+1)\theta - 1}{x \operatorname{cis} \theta - 1} = \frac{x^{n+1} \cos(n+1)\theta - 1 + ix^{n+1} \sin(n+1)\theta}{x \cos \theta - 1 + ix \sin \theta} \quad \checkmark$$

$$= \frac{x^{n+1} \cos(n+1)\theta - 1 + ix^{n+1} \sin(n+1)\theta}{x \cos \theta - 1 + ix \sin \theta} \times \frac{x \cos \theta - 1 - ix \sin \theta}{x \cos \theta - 1 - ix \sin \theta}$$

$$= \frac{a + ib}{a + ib} = \frac{a + ib}{(x \cos \theta - 1)^2 + x^2 \sin^2 \theta} = \frac{a + ib}{x^2 \cos^2 \theta - 2x \cos \theta + 1 + x^2 \sin^2 \theta}$$

$$= \frac{a + ib}{x^2 - 2x \cos \theta + 1}$$

So $\operatorname{Re} \left(\frac{z^{n+1} - 1}{z - 1} \right) = \operatorname{Re} \left(\frac{a + ib}{x^2 - 2x \cos \theta + 1} \right) = \frac{a}{x^2 - 2x \cos \theta + 1}$

From above

$$\frac{z^{n+1} - 1}{z - 1} = \frac{a + ib}{x^2 - 2x \cos \theta + 1}$$

So $a = (x \cos \theta - 1)[x^{n+1} \cos(n+1)\theta - 1] + x^{n+1} \sin(n+1)\theta(x \sin \theta)$

$$= x^{n+2} \cos \theta \cos(n+1)\theta - x^{n+1} \cos(n+1)\theta - (x \cos \theta - 1) + x^{n+2} \sin(n+1)\theta \sin \theta$$

$$= 1 - x \cos \theta + x^{n+2} \cos \theta \cos(n+1)\theta + x^{n+2} \sin(n+1)\theta \sin \theta - x^{n+1} \cos(n+1)\theta$$

$$= 1 - x \cos \theta + x^{n+2} \cos n\theta - x^{n+1} \cos(n+1)\theta \quad \checkmark$$

$$\therefore 1 + x \cos \theta + x^2 \cos 2\theta + \dots + x^n \cos n\theta = \frac{1 - x \cos \theta + x^{n+2} \cos n\theta - x^{n+1} \cos(n+1)\theta}{1 - 2x \cos \theta + x^2} \quad \checkmark$$

Q8-3

$$P(x) = x^3 - 4x^2 + 7x - 6$$

new $P(x) = 8 - 16 + 14 - 6 = 0$

 $x - 2$ is a factor.

	1	-4	7	-6
2	1	-2	-7	6
		2	3	0

$$P(x) = (x - 2)(x^2 - 2x + 3)$$

$$= (x - 2)(x - 1 - i)(x - 1 + i)$$

(3)

b. Let $f(x) = ax^4 + bx^3 + cx^2 + dx + e$

new $f'(x) = 4ax^3 + 3bx^2 + 2cx + d$

new $f'(0) = 0$

$$\therefore d = 0$$

$$f(0) = 1$$

$$\therefore e = 1$$

$$f(1) = 1$$

$$\therefore a + b + c = 0 \quad - A$$

$$f(2) = 13$$

$$\therefore 8a + 12b + 4c = 0 \quad - B$$

$$f(3) = 73$$

$$\therefore 81a + 81b + 27c = 73 \quad - C$$

Solving (A), (B), (C)

$$a = 1, b = 0, c = -1, d = 0, e = 1$$

$$\therefore P(x) = x^4 - x^3 + 1$$

(c) (i) $P(x) = (x - 2)^m Q(x)$

$$P'(x) = m(x - 2)^{m-1} Q(x) + (x - 2)^m Q'(x)$$

$$= (x - 2)^{m-1} [mQ(x) + (x - 2)Q'(x)]$$

$$= (x - 2)^{m-1} P'(x) \quad \text{Q.E.D.}$$

(ii) Let $P(x) = 5x^5 - 3x^3 + c = 0$

Deriv. pro. repeated

$$P'(x) = 25x^4 - 9x^2 = 0$$

$$\therefore x = 0 \text{ or } x = \pm \frac{5}{3}$$

Clearly $x = \frac{5}{3}$.

$$\therefore 5 \left(\frac{5}{3} \right)^5 - 3 \left(\frac{5}{3} \right)^3 + c = 0$$

$$c = \frac{81}{125} - \frac{25}{27}$$

$$c = \frac{162}{625}$$

(d) Given $x^3 + px + q = 0$

Let $y = \frac{1}{x} \Rightarrow x = \frac{1}{y}$ in (d)

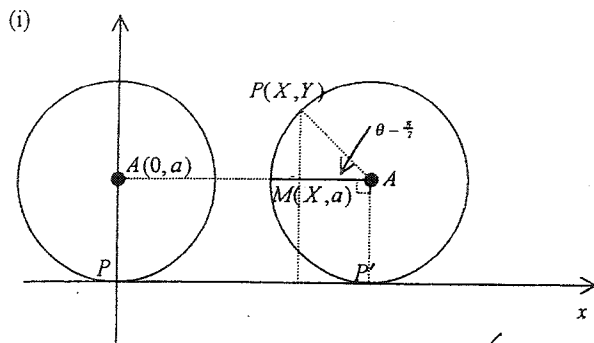
$$\left(\frac{1}{y} \right)^3 + p \cdot \frac{1}{y} + q = 0$$

$$1 + py^2 + qy^3 = 0$$

OR

$$qy^3 + py^2 + 1 = 0 \quad (e)$$

$\therefore$ coefficients are $q, p, 0, 1$.



On the x-axis $PP' = a\theta$ since there is no slipping
 Let $\angle PAP' = \theta \Rightarrow \angle PAM = \theta - \frac{\pi}{2}$

$$Y = a + a \sin(\theta - \frac{\pi}{2}) = a - a \sin(\frac{\pi}{2} - \theta) = a(1 - \cos\theta)$$

$$X = a\theta - a \cos(\frac{\pi}{2} - \theta) = a\theta - a \cos(\theta - \frac{\pi}{2}) = a(\theta - \sin\theta)$$

$$\begin{aligned} \sin(90^\circ - A) &= \cos A \\ \cos(90^\circ - A) &= \sin A \\ \sin(-A) &= -\sin A \\ \cos(-A) &= \cos A \end{aligned}$$

$$x = a(\theta - \sin\theta)$$

$$y = a(1 - \cos\theta)$$

$$\frac{dx}{d\theta} = a(1 - \cos\theta)$$

$$\frac{dy}{d\theta} = a \sin\theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a \sin\theta}{a(1 - \cos\theta)} = \frac{\sin\theta}{1 - \cos\theta}$$

$$\begin{aligned} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} &= \sqrt{1 + \left(\frac{\sin\theta}{1 - \cos\theta}\right)^2} = \sqrt{\frac{(1 - \cos\theta)^2 + \sin^2\theta}{(1 - \cos\theta)^2}} \\ &= \sqrt{\frac{1 - 2\cos\theta + \cos^2\theta + \sin^2\theta}{(1 - \cos\theta)^2}} \\ &= \sqrt{\frac{2 - 2\cos\theta}{(1 - \cos\theta)^2}} = \sqrt{\frac{2(1 - \cos\theta)}{(1 - \cos\theta)^2}} = \frac{\sqrt{2}}{\sqrt{1 - \cos\theta}} \end{aligned}$$

We need to use the following substitution

$$x = a(\theta - \sin\theta)$$

$$dx = a(1 - \cos\theta)d\theta$$

$$0 \leq \theta \leq 2\pi$$

Q8-1

25

i) $\downarrow$ $\begin{aligned} m\ddot{x} &= mg - mkv \\ \ddot{x} &= g - kv \end{aligned}$

ii) $\begin{aligned} \frac{dv}{dt} &= g - kv \\ \frac{dv}{v - \frac{g}{k}} &= -k dt \\ t &= -\frac{1}{k} \ln(g - kv) + C \end{aligned}$

at $t=0, v=0 \Rightarrow C = \frac{1}{k} \ln g$
 $-kt = \ln\left[\frac{g - kv}{g}\right]$
 $e^{-kt} = \frac{g - kv}{g}$
 $g - kv = g e^{-kt}$
 $v = \frac{g}{k}(1 - e^{-kt})$

terminal velocity - net accel. in zero
 max. dist. in $t \rightarrow \infty$
 $\frac{v}{k} = \frac{g}{k}$

ii)

$\downarrow$ $\begin{aligned} m\ddot{y} &= -mg - mkv \\ \ddot{y} &= -g - kv \\ \frac{dy}{dt} &= -\frac{1}{g + kv} dy \end{aligned}$

$t = -\frac{1}{k} \ln(g + kv) + C$
 at $t=0, y=0 \Rightarrow C = \frac{1}{k} \ln g$
 $t = -\frac{1}{k} \ln\left[\frac{g + kv}{g}\right]$

Q. when to rest $v=0$

$t = -\frac{1}{k} \ln\left(\frac{g}{g}\right) = 0$
 $\infty \frac{1}{k} \ln\left(\frac{g}{g}\right)$

(c) $V = \frac{g}{k}(1 - e^{-kt})$
 $t = \frac{1}{k} \ln\left(\frac{g}{g - kv}\right)$
 $= \frac{1}{k} \ln\left(1 - e^{-\frac{g}{g - kv}kt}\right)$
 $= \frac{1}{k} \ln\left(1 - \frac{g}{g - kv}\right)$
 $= \frac{1}{k} \ln\left(\frac{kv}{g - kv}\right)$

$$V = \frac{g}{g - kv}$$

$$V = \frac{g}{g - kv}$$

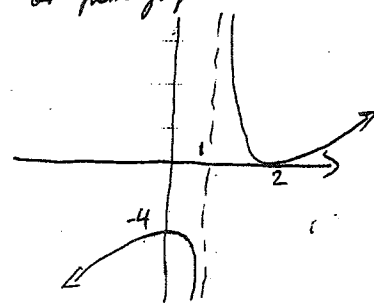
$$V = \frac{V + u}{1 + kv}$$

b) $2z^3$
 $= 2r^3(\cos 3\theta)$
 $= 2r^3 \cos 3\theta$
 $2r^3 \cos 3\theta =$
 $\tan 3\theta =$
 $= \frac{\sqrt{3}}{3}$
 $3\theta = \frac{\pi}{6}$
 $\theta = \frac{\pi}{18}$

at $\theta = \frac{\pi}{18}$
 $2r^3 \sin \frac{\pi}{6} =$
 $r^3 = \frac{3\sqrt{3}}{2}$
 $r = \sqrt[3]{\frac{3\sqrt{3}}{2}}$
 $r = \left(\frac{3}{2}\right)^{\frac{1}{3}}$
 $r = 3^{\frac{1}{6}}$

Q6 let $y = \frac{(x-2)^2}{x-1}$
 Release a particle
 in x for Δ .

or from graph



above y does not
 cross between 0 & 4

(a) $\frac{(x-2)^2}{x-1} = \frac{k}{x}$
 Draw $y = \frac{k}{x}$ and
 $y = \frac{k}{x}$ on above
 graph $y = \frac{k}{x}$ will

c) $V = \pi \int_0^2 x dx$
 $= 12\pi$

d) $\sqrt{n^2 + n} \times \frac{1}{n}$
 $= \frac{n^2 + n}{\sqrt{n^2 + n}}$
 $= \frac{n}{\sqrt{n^2 + n}}$
 $= \frac{1}{\sqrt{1 + \frac{1}{n}}}$
 $= \frac{1}{\sqrt{1 + \frac{1}{n}}}$
 as $n \rightarrow \infty$
 $\lim = \frac{1}{\sqrt{1 + 0}} = 1$